

Pranshu Gupta

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SUMMARY

- AI infrastructure engineer with 9 years of experience, leading LLM inference and orchestration for Microsoft Copilot at global scale. Shipped inference optimizations with sustained GPU savings, significantly reducing operational costs.

PROFESSIONAL EXPERIENCE

- **Microsoft AI** / *Member of Technical Staff / Redmond, WA* *Aug 2025 – Present*
 - Reduced production GPU footprint by **1,320 H100 GPUs** by analyzing the inference pipeline and eliminating non-essential classifiers and summarizers on the GPT-5.1 instant path.
 - Led end-to-end rollout of **GPT-5.1 instant and reasoning models** to production by designing prompt templates, constrained-sampling, chain of thought handling, and safety integrations across six partner teams; owned launch-readiness evaluations with the Response Quality squad, including **30% improvement in personality evals**.
 - Designed and shipped **Adaptive Thinking**, a dynamic system that adjusts the thinking effort per message based on intent/capability classification to control reasoning token usage.
 - Architected a request-scoped **feature flag system** for the inference API, adopted by MAI and M365 as the standard mechanism for A/B experiments and incremental rollouts.
 - Unblocked GA of a new LLM input safety classifier, helping reduce model ship cycle from **>4 months to <1 month**.
 - Improved session-management reliability by fixing affinity-drain ordering, eliminating deploy-time connection failures, and reducing **Redis write volume by 9%** via session-affinity deduplication.
- **Microsoft Azure AI** / *Senior Software Engineer / Redmond, WA* *Jan 2025 – Aug 2025*
 - Designed a distributed **token-based rate limiter** using Tiktoken for precise token counting, replacing a length-based estimator with **219% average error**; implemented pub-sub quota state synchronization across instances.
 - Contributed to an intelligent **model router** for per-request model selection based on cost/complexity signals, reducing per-inference cost by **55%**.
- **Microsoft Azure Core** / *Senior Software Engineer / Redmond, WA* *Mar 2024 – Dec 2024*
 - Designed and shipped a token-based job throttling system with multi-bucket quota enforcement and retry-storm protection, preventing partition-level noisy-neighbor throttling in background-job infrastructure of Azure Control Plane.
 - Single-handedly designed and shipped async operation URI signing, adopted by **242 Azure service teams** and enforced for **220+ resource providers**.
- **Microsoft Azure Core** / *Software Engineer II / Redmond, WA* *Jun 2020 – Mar 2024*
 - Designed and shipped async operation callbacks replacing polling, reducing ARM-to-resource-provider traffic by up to **94%** (130M fewer requests/day) and improving p95 latency by **12 seconds** for Azure Container Instances.
 - Implemented exponential backoff for resource-consistency jobs, cutting executions by **70% (700M/day)**.
 - Owned bulk resource delete API end-to-end (dependency resolution, async scheduling, staged retries), reducing **P90 latency by 50%** and **P99 by 65%**.
- **Microsoft Core Services** / *Software Engineer / Hyderabad, India* *Jun 2017 – Aug 2019*
 - Migrated customer data enrichment service to serverless design, reducing cost by **90%** and improving throughput by **3x**.

TECHNICAL SKILLS

- **AI/ML & Inference:** LLM inference/serving, prompt engineering, model evaluations, safety orchestration, RAG
- **Languages & Systems:** Python, C#, C++, KQL, OpenAI API, Kubernetes, Redis, Cosmos DB, Service Bus, .NET

TECHNICAL WRITING

- *Runtime Optimizations for LLMs* (2026). Production notes on model offloading, KV cache management, and quantization for latency/cost-efficient LLM serving. pranshug.com/blog/runtime-optimizations-for-llms

PUBLICATIONS

- Saha, K., **Gupta, P.** et al. *Observer Effect in Social Media Use*. ACM CHI, 2024. doi:10.1145/3613904.3642078
- El Sherief, M., Saha, K., **Gupta, P.** et al. *Impacts of school shooter drills on the psychological well-being of American K-12 school communities*. Nature Humanities and Social Sciences, 2021. [nature.com/articles/s41599-021-00993-6](https://www.nature.com/articles/s41599-021-00993-6)

EDUCATION

- **Georgia Institute of Technology** / *M.S. Computer Science (Machine Learning) / Atlanta, GA* *May 2020*
- **Indian Institute of Technology, Kanpur** / *B.Tech, C.S. and Eng. / Kanpur, India* *May 2017*